

Appendix for Property Enhanced Instruction Tuning for Multi-task Molecule Generation with Large Language Models

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A Analysis on Hyperparameters

We further analyzed the impact of hyperparameter settings by varying the ratio of α to β to examine how different configurations affect model performance. As shown in Table 1, we observed that setting $\alpha:\beta$ to 1:5 yields the best results. Consequently, we adopt the 1:5 configuration following the setting used in SPMM [Chang and Ye, 2024].

B Template Filling

We show the templates in Figure 1 for synthesizing instruction data.

C Details of Property Prediction Tasks

Following SPMM [Chang and Ye, 2024], we adopt four commonly-used binary property prediction tasks to evaluate the performance of PEIT-GEN, including BBBP, BACE, Clintox, and SIDER dataset. The BBBP dataset contains 2,050 molecular samples and aims to predict whether these molecules can cross the blood-brain barrier. The BACE dataset includes 1,513 molecular samples and is used to predict whether a molecule can inhibit the activity of the BACE1 enzyme. The Clintox dataset contains 1,478 molecular samples and is primarily used to predict the toxicity of compounds. The SIDER dataset consists of 1,427 drug samples and is used to predict whether a drug will cause specific side effects. Specifically, we use scaffold splitting and each dataset is divided into a training set, validation set, and test set in a ratio of 8:1:1, respectively.

D Details of Baselines

We compare our model against a variety of baselines which can be categorized as follows:

Baselines on molecule captioning task:

MolT5 [Edwards *et al.*, 2022] is a framework for pre-training models on unlabeled text and molecular data. It introduces tasks like molecule captioning and generating molecules from text.

BioT5 [Pei *et al.*, 2023] is a biology-focused pre-trained language model trained on diverse biological data, linking text with molecular and protein information.

MolXPT [Liu *et al.*, 2023a] is a pre-trained language model for molecular science that enriches both text and molecular

Model	$\alpha : \beta$	BLEU $\uparrow$	METEOR $\uparrow$	ROUGE $\uparrow$	R^2 $\uparrow$	RMSE $\downarrow$
PEIT-GEN	1:1	0.582	0.659	0.691	0.893	0.219
PEIT-GEN	1:3	0.571	0.639	0.686	0.903	0.193
PEIT-GEN	1:5	0.598	0.676	0.700	0.910	0.169

Table 1: The impact of hyperparameters α and β on the results is presented, where BLEU, METEOR, and ROUGE are evaluation metrics on the molecule captioning task, while R^2 and RMSE are used to assess performance on the molecular property prediction task.

SMILES representations by replacing molecular names in the text with SMILES notation.

GIT-Mol [Liu *et al.*, 2024] is a multi-modal LLM designed for molecular science, integrating graph, image, and text data. It performs well in tasks like molecule captioning, text-to-molecule generation, image recognition, and property prediction.

MolCA [Liu *et al.*, 2023b] is a model that combines molecular graphs with textual descriptions, excelling in molecular representation learning, cross-modal reasoning, and tasks such as property prediction, generation, and interaction.

Text+Chem-T5 [Christofidellis *et al.*, 2023] is a multimodal model based on the T5 architecture, specifically designed for joint chemistry-text tasks. By integrating chemical data with natural language text, it enhances performance in chemical text understanding, molecular property prediction, and reaction generation tasks.

Baselines on molecular property prediction:

SPMM [Chang and Ye, 2024] is a multi-modal molecular pre-trained model that combines molecular structure information and biochemical properties by aligning two distinct features into a shared embedding space.

D-MPNN [Yang and others, 2019] D-MPNN is specifically designed for processing molecular graph data. It efficiently captures atomic interactions and chemical bond information through a directed message-passing mechanism, providing strong support for molecular property prediction.

N-GramRF [Liu *et al.*, 2019] extracts N-Gram features from molecular sequences and integrates them with a Random Forest [Breiman, 2001] model to capture local structural information of molecules. It is suitable for molecular property prediction tasks, offering strong robustness and easy implementation.

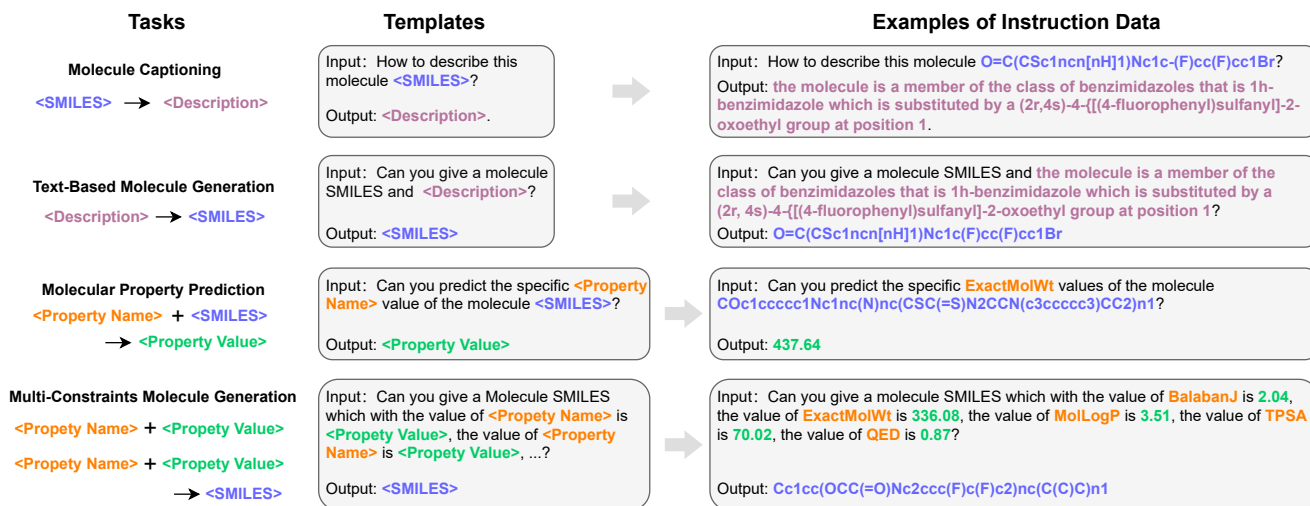


Figure 1: Examples of template filling with unstructured data according to four different downstream tasks for obtaining a variety of instruction data for supervised fine-tuning large language models.

N-GramXGB [Liu *et al.*, 2019] also utilizes N-Gram features but employs the XGBoost [Chen and Guestrin, 2016] model for prediction. It efficiently handles high-dimensional data and captures nonlinear relationships, often outperforming Random Forest in predictive performance.

PretrainGNN [Hu *et al.*, 2019] performs pre-training on molecular graph-structured data through self-supervised learning tasks, thereby learning universal representations of nodes and edges within the graph. This significantly enhances the model’s performance in molecular property prediction tasks.

GROVER [Rong *et al.*, 2020] leverages multiple self-supervised learning tasks to learn universal representations of atoms and bonds in molecular structures, significantly enhancing performance in downstream tasks such as molecular property prediction and drug discovery.

ChemRL-GEM [Fang *et al.*, 2022] employs Graph Neural Networks (GNNs) to learn the embedding representations of molecular graphs and utilizes reinforcement learning to optimize these representations, thereby better accomplishing tasks such as molecular property prediction and molecular generation.

ChemBERTa [Ahmad *et al.*, 2022] is pre-trained on a large-scale chemical literature and biomedical corpora, learning linguistic features specific to the chemistry and biomedical domains. This enables it to excel in tasks such as molecular property prediction, drug discovery, and biomedical text mining.

MolFormer [Ross *et al.*, 2022] captures global atomic interactions within molecules using self-attention and learns universal molecular representations through pretraining on large-scale datasets, demonstrating strong performance in property prediction and molecular generation tasks.

Baselines of LLMs:

LLaMa3 [Touvron *et al.*, 2023] is an open-source LLM, suitable for various NLP tasks such as summarization, question answering, and translation.

LLaMa3.1 [Dubey *et al.*, 2024] is a series of updated open-source LLM based on LLaMa3, featuring a stronger parameter scale and higher performance.

Qwen2.5 [Yang *et al.*, 2024] is an open-source large model that has been pre-trained on a dataset containing 18 trillion tokens. It has achieved significant improvements in overall capabilities and excels in a wide range of NLP tasks.

GPT-3.5 Turbo [OpenAI, 2023] is an advanced large language model developed by OpenAI, optimized for efficient inference and versatile natural language understanding and generation tasks. Built upon the transformer architecture, GPT-3.5 Turbo demonstrates strong performance across a wide range of NLP benchmarks, including text completion, summarization, translation, and dialogue systems. Its design balances high accuracy with reduced computational cost, making it suitable for scalable real-world applications.

Mol-Instructions [Fang *et al.*, 2023] is a natural language instruction dataset for biomolecules, designed to enhance the capabilities of large-scale pre-trained models in the biomolecular domain. This dataset combines biomolecules (such as proteins, DNA, RNA, etc.) with natural language instructions, supporting tasks such as molecule generation, molecule modification, and reaction prediction. We use the LLaMa3.1-8B model after SFT on this instruction dataset.

BioMedGPT [Zhang *et al.*, 2024] is a multimodal pre-trained model for the biomedical field, leveraging self-supervised learning and cross-modal alignment to learn universal representations from large-scale data, excelling in text understanding, medical image analysis, and molecular property prediction.

InstructMol-GS [Cao *et al.*, 2023] is an instruction-tuned molecular generation model that maps natural language to molecular structures, enabling targeted molecule design and demonstrating strong generative capabilities in drug discovery and materials science.

MolReGPT [Li *et al.*, 2024] is a molecule-text translation framework based on LLMs. It utilizes a molecular similar-

Model	Modality	Data Size ↓	R^2 ↑	RMSE ↓
SPMM [Chang and Ye, 2024]	$S, \mathcal{P}$	1.5M	0.921	0.194
PEIT-GEN (Ours)	$S, \mathcal{P}, \mathcal{T}$	480K	0.910	0.169

Table 2: Comparing performance of our PEIT-GEN to SPMM on molecular property prediction.

ity retrieval mechanism to select examples, enabling efficient molecule generation and understanding without fine-tuning.

ChemDFM [Zhao *et al.*, 2025] is a large language foundation model for the field of chemistry. Trained on 34 billion tokens from chemical literature and 2.7 million instructions, it demonstrates strong capabilities in understanding and reasoning about chemical knowledge. It supports tasks such as molecule recognition, design, property prediction, and reaction analysis, outperforming many open-source large language models.

ChatGLM [GLM *et al.*, 2024] is an open-source bilingual large language model optimized for Chinese and English. It supports instruction tuning and multi-round dialogue, making it adaptable to domain-specific tasks such as molecular captioning and property prediction with appropriate prompting.

Galactica [Taylor *et al.*, 2022] is a large language model pretrained on scientific texts, including papers, molecules, and protein sequences. Designed to assist scientific reasoning and knowledge retrieval, it supports molecule-related tasks to a limited extent through its exposure to structured scientific data during pretraining.

Vicuna [Chiang *et al.*, 2023] is an open-source language model fine-tuned from LLaMA using user-shared conversations. It focuses on improving instruction-following and dialogue capabilities, and can be adapted to domain-specific tasks through fine-tuning, despite lacking scientific domain pretraining.

E Evaluation Metrics

We evaluated the quality of generated text using BLEU [Papineni *et al.*, 2002], METEOR [Banerjee and Lavie, 2005], and ROUGE scores. These metrics evaluate the similarity between generated texts and reference descriptions, effectively quantifying the accuracy and diversity of the generated descriptions. For the text-based molecule generation task, we further use molecular fingerprints (FTS) [Cereto-Massagué *et al.*, 2015] and validity measures to assess molecular similarity and validity, including Validity, Levenshtein [Levenshtein, 1966], MACCS FTS, Morgan FTS, and RDKit FTS [Landrum and others, 2013]. For the task of molecular property prediction, we chose to use the commonly used RMSE to measure the difference between the predicted values and the molecular property values calculated by RDKit for comparison, for the experiments on MoleculeNet, we use AUC-ROC to evaluate the accuracy for property prediction tasks. In the case of multi-constraint molecule generation, in addition to RMSE, we also employed R^2 to assess the accuracy of the generated molecules.

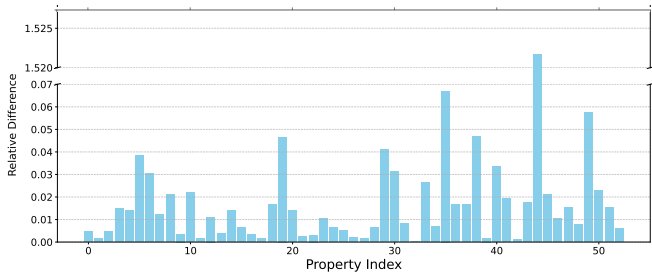


Figure 2: The relative difference represent the variation between the PEIT-GEN predicted values and the actual values for 53 distinct molecular properties.

F Molecular Property Prediction

Following SPMM [Chang and Ye, 2024], we further compare PEIT-GEN with SPMM on external dataset. The comparison result on molecular property prediction is shown in Table 2. Specifically, we randomly sample 1,000 molecules from the ZINC dataset which are not included in the training set. Compared to SPMM that is specifically designed for property prediction, PEIT-GEN achieves comparable performance while using only one-third of the data size across three modalities. We found that PEIT-GEN outperformed SPMM in terms of RMSE, while SPMM was slightly ahead by 0.11% on R^2 metric. These results demonstrate that PEIT-GEN can generate high-quality biochemical properties of molecules, highlighting the critical role of high-quality multi-modal data in advancing molecular understanding tasks. To further illustrate the predictive performance of PEIT-GEN on the 53 molecular properties, we calculated the relative difference of these properties, as shown in Figure 2.

G Ablation Study on Molecule Property Prediction

To demonstrate the effectiveness of the cross-modal matching loss $\mathcal{L}_{\text{match}}$ and cross-modal contrastive loss $\mathcal{L}_{\text{contrastive}}$ within the PEIT-GEN model for molecular property prediction, we conducted ablation experiments analogous to those performed in the molecule captioning task, as shown in Table 3. The results confirm that both $\mathcal{L}_{\text{match}}$ and $\mathcal{L}_{\text{contrastive}}$ contribute significantly to cross-modal feature alignment, thereby improving the model’s predictive performance.

Metric	PEIT-GEN	w/o $\mathcal{L}_{\text{match}}^{\text{st}}$	w/o $\mathcal{L}_{\text{contrastive}}^{\text{st}}$	w/o $\mathcal{L}_{\text{contrastive}}^{\text{sp}}$	w/o $\mathcal{L}_{\text{match}}^{\text{sp}}$
R^2 ↑	0.910	0.788	0.772	0.572	0.493
RMSE ↓	0.169	0.192	0.242	0.442	0.381

Table 3: Ablation study on the impact of different training objectives on molecular property prediction.

H Case Study

As for qualitative analysis, we show some case study for different tasks, directly comparing the response by LLaMa3.1 and our PEIT-LLM.

Table 4 shows the examples for molecule captioning. From the responses, we can see that the captions generated by our

model are closer to the ground truth captions, whereas the original model often provides vague answers. This demonstrates that our model has a better understanding of molecules and can generate high-quality responses to complete molecular annotation tasks.

Table 5 shows the examples for text-based molecule generation. We can see that, compared with LLaMa3.1, the molecules generated by our model are more structurally reasonable and similar to the ground truth.

Table 6 shows the examples for molecular property prediction. LLaMa3.1 performs poorly in predicting specific molecular property and lacks the ability to accurately predict molecular property values. However, after instruction fine-tuning, our PEIT-LLM demonstrates excellent performance in predicting molecular property values, providing a strong foundation for its outstanding performance in multi-constraint molecule generation tasks.

Table 7 shows the examples for challenging multi-constraint molecule generation task. It can be observed that our model generated molecule with close properties according to multiple constraints. This indicates that the multi-task instruction data generated by our pre-trained model can enhance the capabilities of LLMs for such challenging task.

References

[Ahmad *et al.*, 2022] Walid Ahmad, Elana Simon, Seyone Chithrananda, Gabriel Grand, and Bharath Ramsundar. Chemberta-2: Towards chemical foundation models. *arXiv preprint arXiv:2209.01712*, 2022.

[Banerjee and Lavie, 2005] Satanejeev Banerjee and Alon Lavie. METEOR: An automatic metric for mt evaluation with improved correlation with human judgments. In *Proceedings of the acl workshop on intrinsic and extrinsic evaluation measures for machine translation and/or summarization*, pages 65–72, 2005.

[Breiman, 2001] Leo Breiman. Random forests. *Machine learning*, 45:5–32, 2001.

[Cao *et al.*, 2023] He Cao, Zijing Liu, Xingyu Lu, Yuan Yao, and Yu Li. Instructmol: Multi-modal integration for building a versatile and reliable molecular assistant in drug discovery, 2023.

[Cereto-Massagué *et al.*, 2015] Adrià Cereto-Massagué, María José Ojeda, Cristina Valls, Miquel Mulero, Santiago Garcia-Vallvé, and Gerard Pujadas. Molecular fingerprint similarity search in virtual screening. *Methods*, 71:58–63, 2015.

[Chang and Ye, 2024] Jinho Chang and Jong Chul Ye. Bidirectional generation of structure and properties through a single molecular foundation model. *Nature Communications*, 15(1):2323, 2024.

[Chen and Guestrin, 2016] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, pages 785–794, 2016.

[Chiang *et al.*, 2023] Zhihao Chiang, Zhuohan Zhu, Xinyang Zeng, Yian Xu, Zi Lin, Hao Li, Lianmin Zhang,

Shreyas Jain, Rohan Taori, Ishaan Dave, et al. Vicuna: An open-source chatbot impressing gpt-4 with 90%* chatgpt quality, 2023.

[Christofidellis *et al.*, 2023] Dimitrios Christofidellis, Giorgio Giannone, Jannis Born, Ole Winther, Teodoro Laino, and Matteo Manica. Unifying molecular and textual representations via multi-task language modelling. In *Proceedings of the 40th International Conference on Machine Learning*, pages 6140–6157, 2023.

[Dubey *et al.*, 2024] Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

[Edwards *et al.*, 2022] Carl Edwards, Tuan Lai, Kevin Ros, Garrett Honke, Kyunghyun Cho, and Heng Ji. Translation between molecules and natural language. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 375–413, 2022.

[Fang *et al.*, 2022] Xiaomin Fang, Lihang Liu, Jieqiong Lei, Donglong He, Shanzhuo Zhang, Jingbo Zhou, Fan Wang, Hua Wu, and Haifeng Wang. Geometry-enhanced molecular representation learning for property prediction. *Nature Machine Intelligence*, 4(2):127–134, 2022.

[Fang *et al.*, 2023] Yin Fang, Xiaozhuan Liang, Ningyu Zhang, Kangwei Liu, Rui Huang, Zhuo Chen, Xiaohui Fan, and Huajun Chen. Mol-Instructions: A large-scale biomolecular instruction dataset for large language models. In *International Conference on Learning Representations*, 2023.

[GLM *et al.*, 2024] Team GLM, Aohan Zeng, Bin Xu, Bowen Wang, Chenhui Zhang, Da Yin, Dan Zhang, Diego Rojas, Guanyu Feng, Hanlin Zhao, et al. Chatglm: A family of large language models from glm-130b to glm-4 all tools. *arXiv preprint arXiv:2406.12793*, 2024.

[Hu *et al.*, 2019] Weihua Hu, Bowen Liu, Joseph Gomes, Marinka Zitnik, Percy Liang, Vijay Pande, and Jure Leskovec. Strategies for pre-training graph neural networks. *arXiv preprint arXiv:1905.12265*, 2019.

[Landrum and others, 2013] Greg Landrum et al. Rdkit: A software suite for cheminformatics, computational chemistry, and predictive modeling. *Greg Landrum*, 8(31.10):5281, 2013.

[Levenshtein, 1966] VI Levenshtein. Binary codes capable of correcting deletions, insertions, and reversals. *Proceedings of the Soviet physics doklady*, 1966.

[Li *et al.*, 2024] Jiatong Li, Yunqing Liu, Wenqi Fan, Xiaoyong Wei, Hui Liu, Jiliang Tang, and Qing Li. Empowering molecule discovery for molecule-caption translation with large language models: A chatgpt perspective. *IEEE transactions on knowledge and data engineering*, 2024.

[Liu *et al.*, 2019] Shengchao Liu, Mehmet F Demirel, and Yingyu Liang. N-gram graph: Simple unsupervised representation for graphs, with applications to molecules. *Advances in neural information processing systems*, 32, 2019.

- [Liu *et al.*, 2023a] Zequn Liu, Wei Zhang, Yingce Xia, Lijun Wu, Shufang Xie, Tao Qin, Ming Zhang, and Tie-Yan Liu. MolXPT: Wrapping molecules with text for generative pre-training. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 1606–1616, 2023.
- [Liu *et al.*, 2023b] Zhiyuan Liu, Sihang Li, Yanchen Luo, Hao Fei, Yixin Cao, Kenji Kawaguchi, Xiang Wang, and Tat-Seng Chua. Molca: Molecular graph-language modeling with cross-modal projector and uni-modal adapter. In *EMNLP*, 2023.
- [Liu *et al.*, 2024] Pengfei Liu, Yiming Ren, Jun Tao, and Zhixiang Ren. Git-mol: A multi-modal large language model for molecular science with graph, image, and text. *Computers in biology and medicine*, 171:108073, 2024.
- [OpenAI, 2023] OpenAI. GPT-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- [Papineni *et al.*, 2002] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, pages 311–318, 2002.
- [Pei *et al.*, 2023] Qizhi Pei, Wei Zhang, Jinhua Zhu, Kehan Wu, Kaiyuan Gao, Lijun Wu, Yingce Xia, and Rui Yan. BioT5: Enriching cross-modal integration in biology with chemical knowledge and natural language associations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 1102–1123, 2023.
- [Rong *et al.*, 2020] Yu Rong, Yatao Bian, Tingyang Xu, Weiyang Xie, Ying Wei, Wenbing Huang, and Junzhou Huang. Self-supervised graph transformer on large-scale molecular data. *Advances in neural information processing systems*, 33:12559–12571, 2020.
- [Ross *et al.*, 2022] Jerret Ross, Brian Belgodere, Vijil Chen-thamarakshan, Inkit Padhi, Youssef Mroueh, and Payel Das. Large-scale chemical language representations capture molecular structure and properties. *Nature Machine Intelligence*, 4(12):1256–1264, 2022.
- [Taylor *et al.*, 2022] Ross Taylor, Marcin Kardas, Guillem Cucurull, Thomas Scialom, Anthony Hartshorn, Elvis Saravia, Andrew Poulton, Viktor Kerkez, and Robert Stojnic. Galactica: A large language model for science. *arXiv preprint arXiv:2211.09085*, 2022.
- [Touvron *et al.*, 2023] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.
- [Yang and others, 2019] K. Yang et al. Analyzing learned molecular representations for property prediction. *Journal of Chemical Information and Modeling*, 59(8):3370–3388, 2019.
- [Yang *et al.*, 2024] An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, Chengpeng Li, Chengyuan Li, Dayiheng Liu, Fei Huang, et al. Qwen2 technical report. *arXiv preprint arXiv:2407.10671*, 2024.
- [Zhang *et al.*, 2024] Kai Zhang, Rong Zhou, Eashan Adhikarla, Zhiling Yan, Yixin Liu, Jun Yu, Zhengliang Liu, Xun Chen, Brian D Davison, Hui Ren, et al. A generalist vision-language foundation model for diverse biomedical tasks. *Nature Medicine*, pages 1–13, 2024.
- [Zhao *et al.*, 2025] Zihan Zhao, Da Ma, Lu Chen, Liangtai Sun, Zihao Li, Yi Xia, Bo Chen, Hongshen Xu, Zichen Zhu, Su Zhu, et al. Developing chemdfm as a large language foundation model for chemistry. *Cell Reports Physical Science*, 6(4), 2025.

Input for Molecule Captioning	LLaMa3.1 Output	Our PEIT-LLM Output	Ground Truth
1. How to describe this molecule <chem>CCCCC1OC1C=CC(O)CC=C</chem> <chem>CC=CCCC(=O)[O-]</chem> ?	This is a complex organic molecule containing a cyclic ether, an alkene, an alcohol, a ketone, and an enolate anion.	The molecule is a prostaglandin carboxylic acid anion that is the conjugate base of prostaglandin F2alpha, obtained by deprotonation of the carboxy group; major species at pH 7.3 . It is a conjugate base of a prostaglandin F2alpha.	The molecule is an epoxy(hydroxy)icosatrienoate that is the conjugate base of 11 hydroxy-(14R,15S)-epoxy-(5Z,8Z,12E)-icosatrienoic acid, obtained by deprotonation of the carboxy group; major species at pH 7.3 . It is a conjugate base of an 11 hydroxy-(14R,15S)-epoxy-(5Z,8Z,12E)-icosatrienoic acid.
2. How to describe this molecule [Te]?	The molecule [Te] represents a tellurium atom in its elemental form.	The molecule is the stable isotope of tellurium with relative atomic mass 127.904 , 7.00 atom percent natural abundance and nuclear spin 1/2 .	The molecule is the stable isotope of tellurium with relative atomic mass 124.904425 , 71.4 atom percent natural abundance and nuclear spin 1/2 .
3. How to describe this molecule <chem>CN(N=O)C(N)=O</chem> ?	This molecule is a nitrosoguanidine, characterized by a central carbon atom double-bonded to nitrogen and oxygen, with an additional nitrogen atom attached to the carbon.	The molecule is a nitrosamine that is urea in which one of the nitrogens is substituted by methyl and nitroso groups. It has a role as a carcinogenic agent , a mutagen , a teratogenic agent and an alkylating agent .	The molecule is a member of the class of N-nitrosoureas that is urea in which one of the nitrogens is substituted by methyl and nitroso groups. It has a role as a carcinogenic agent , a mutagen , a teratogenic agent and an alkylating agent .

Table 4: Examples of responses by LLaMa3.1 and our PEIT-LLM on molecule captioning.

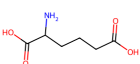
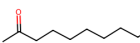
Input for Text-based Molecule Generation	LLaMa3.1 Output	Our PEIT-LLM Output	Ground Truth
1. Can you give a molecule SMILES and the molecule is an amino dicarboxylic acid that is adipic acid in which one of the hydrogens at position 3 is replaced by an amino group. It is a beta-amino acid an amino dicarboxylic acid and a gamma-amino acid. It derives from an adipic acid?	<chem>Cn1c(=O)c2c(nc(NCC3CCCC3)n2CNC(CCCC(=O)O)C(=O)O</chem> <chem>C(=O)c2cccc2)n(C)c1=O</chem>		<chem>C(CC(=O)O)C(CC(=O)O)N</chem>
2. Can you give a molecule SMILES and the molecule is a methyl ketone that is decane in which the methylene hydrogens at position 2 are replaced by an oxo group. It derives from a hydride of a decane?	<chem>CN(C)C1=CC=C2[NH1]C=3C(C([NH&CCCCCCC(=O)CC</chem> <chem>+1])=O)=CC(=O)C(=O)C=3OC2=C1</chem>		<chem>CCCCCCCCC(=O)C</chem>
3. Can you give a molecule SMILES and the molecule is a member of the class of Nitrosoureas that is urea in which one of the nitrogens is substituted by methyl and nitroso groups. It has a role as a carcinogenic agent, a mutagen, a teratogenic agent and an alkylating agent?	<chem>C1=CC(=O)OC2=CC(=C(C=C21)O</chem> <chem>[C@H]3[C@H]([C@H]([C@H]</chem> <chem>([C@H](O3)CO)O)O)O</chem>	<chem>CCOP(=S)(OCC)OC1=NN(C=N1)CCOP(=S)(OCC)OCC1=CC(=O)</chem> <chem>N(C2=CC=CC=C2)N1</chem>	<chem>CCOP(=S)(OCC)OCC1=CC(=O)N1</chem>

Table 5: Examples of responses by LLaMa3.1 and our PEIT-LLM for text-based molecule generation.

Input	LLaMa3.1 Output	Our PEIT-LLM Output	Ground Truth
1. Can you predict the specific MolLogP values of the molecule? <chem>CC1CC(C)CN(S(=O)(=O)c2ccc(C(=O)Nc3nnc(C4CC4)O3)cc2)C1</chem>	23.42	2.65	2.66
2. Can you predict the specific ExactMolWt values of the molecule? <chem>COC1ccc(NC(=O)Cn2ccn(-c3ccc(F)c(F)c3)c(=O)c2=O)cc1C1</chem>	13.77	404.94	404.25
3. Can you predict the specific MolLogP values of the molecule? <chem>Cc1cc2c(cc1)C(=O)NC(C)C2</chem>	18.98	2.53	2.52

Table 6: Examples of responses by LLaMa3.1 and our PEIT-LLM for molecular property prediction.

Input for Multi-Constraint Molecule Generation	LLaMa3.1 Output (Verified Values by RD-Kit)	Our PEIT-LLM Output (Verified Values by RD-Kit)
1. Can you give a molecule SMILES which with the value of BalabanJ is 1.68, the value of ExactMolWt is 492.20, the value of MolLogP is 3.14, the value of TPSA is 138.76, the value of QED is 0.33?	<chem>NN=c1sc2cccc2n1-c1ccccc1</chem> (BalabanJ is 2.42, ExactMolWt is 241.06, MolLogP is 2.46, TPSA is 43.30, QED is 0.51)	<chem>CCOc1cc(C=NNC(=O)c2ccncc2)ccc1OS(=O)(=O)c1ccc(NC(C)=O)cc1</chem> (BalabanJ is 1.74, ExactMolWt is 482.12, MolLogP is 2.97, TPSA is 136.04, QED is 0.27)
2. Can you give a molecule SMILES which with the value of BalabanJ is 1.33, the value of ExactMolWt is 467.20, the value of MolLogP is 3.54, the value of TPSA is 92.42, the value of QED is 0.52?	<chem>O=C(CN1CCN(c2ccc(Cl)cc2)CC1)Nc1ccc(F)cc1</chem> (BalabanJ is 1.49, ExactMolWt is 365.11, MolLogP is 3.37, TPSA is 35.58, QED is 0.90)	<chem>O=C(Cc1ccc(F)cc1)COC(=O)c1ccc(S(=O)(=O)N2CCCc3ccc(F)cc3)cc1</chem> (BalabanJ is 1.39, ExactMolWt is 468.11, MolLogP is 3.76, TPSA is 92.78, QED is 0.55)
3. Can you give a molecule SMILES which with the value of BalabanJ is 1.98, the value of ExactMolWt is 303.13, the value of MolLogP is 3.06, the value of TPSA is 64.11, the value of QED is 0.92?	<chem>O=C(NCc1cccc(F)c1)Nc1nnc(C2CC(O)C2)s1</chem> (BalabanJ is 1.51, ExactMolWt is 368.09, MolLogP is 1.18, TPSA is 116.60, QED is 0.62)	<chem>Cc1ccc(NC(=O)C2CCc3ccc(C)cc3)cc1</chem> (BalabanJ is 2.11, ExactMolWt is 306.10, MolLogP is 2.88, TPSA is 67.43, QED is 0.90)

Table 7: Examples of responses by LLaMa3.1 and our PEIT-LLM for multi-constraint molecule generation, and the verified property values of the output molecule are shown in the brackets.